CANCER EPIGENETICS COURSE

BASICS OF GOOGLE COLAB, R PROGRAMMING LANGUAGE

Presenter: Tran Duc Phuc

21st December 2025

Homework

Homework 1:
1. Vector: Create a numeric vector named grades containing five exam scores. (9, 8, 5, 4, 7)
2. Matrix: Construct a 5x2 matrix called attendance where each row represents a student and columns represent "Days Present" and "Days Absent".
3. List: Create a list named student_profile that stores a student's name (character), their grades vector, and their attendance matrix. (any name)
4. Data Frame: Build a data frame named class_summary that includes columns for Student_Name and Final_Grade (calculated as the mean of a vector).
5. Access the grades vector from within your student_profile list using the $ operator and calculate its average.

Link to google colab: https://colab.research.google.com/drive/1UKARP5im6-lIDUl2risx8qfTIYFwKnz5?usp=sharing

Homework

Homework 2:

1. Vector: Define a character vector of item_names and a numeric vector of unit_prices.
2. Matrix: Create a matrix called store_stock with 3 rows (stores) and 4 columns (products), filling it with random inventory counts.
3. Data Frame: Combine your item_names and unit_prices into a data frame named price_list.
4. List: Create an inventory_packet list that contains the price_list data frame and the store_stock matrix.
5. Add a new column to the price_list data frame called discounted_price by multiplying the unit_prices vector by 0.9.

Link to google colab: https://colab.research.google.com/drive/1UKARP5im6-lIDUI2risx8qfTIYFwKnz5?usp=sharing

Homework

Homework 3:

1. Vector: Create a character vector of five cities and a numeric vector of their populations.
2. Matrix: Build a 5x5 distance matrix where each element m[i,j] represents the distance between two cities.
3. Data Frame: Create a data frame city_info from your cities and populations vectors. Create a second data frame city_coords with cities and their "Latitude/Longitude".
4. List: Combine both data frames and the distance matrix into a master list called region_map.
5. Use the merge() function to combine city_info and city_coords by the "City" column into a single comprehensive data frame

Link to google colab: https://colab.research.google.com/drive/1UKARP5im6-lIDUl2risx8qfTIYFwKnz5?usp=sharing